

LOVASTIN[®] TABLET 20MG

Lovastatin, the active ingredient of Lovastatin Tablets, is a cholesterol-lowering agent. It is converted in vivo by esterases to the corresponding open hydroxyacid which is a potent inhibitor of endogenous cholesterol synthesis.

Ingredient(s):

Each tablet contains:

Lovastatin 20mg

Pharmacodynamic:

1. Lovastatin is a cholesterol-lowering agent isolated from a strain of *A. terreus*. After oral ingestion, lovastatin which is an inactive lactone, is hydrolyzed to the corresponding β -hydroxyacid form. This principal metabolite is a specific inhibitor of 3-hydroxy-3-methylglutaryl-coenzyme A (HMG-CoA) reductase. This enzyme catalyses the conversion of HMG-CoA to mevalonate which is an early and rate limiting step in the biosynthesis of cholesterol.
2. Lovastatin reduces cholesterol production by the liver and induces some changes in cholesterol transport and disposition in the blood and tissues. The mechanism of this effects is believed to involve both reduction of the synthesis of low density lipoprotein (LDL), and in increases in LDL catabolism as a result of induction of the hepatic LDL receptors. The mechanism of the LDL lowering effects of lovastatin may also involve reduction of the very low density lipoprotein cholesterol concentration.

Pharmacokinetics

1. Following an oral dose of ¹⁴C-labeled lovastatin in man, 10% of the dose was excreted in the urine and 83% in the feces. The latter represents the absorbed as well as any unabsorbed drug that is excreted in the bile. As a consequence of extensive hepatic extraction of lovastatin, the availability of drug to the general circulation is low and variable.
2. Both lovastatin and β -hydroxyacid metabolite are highly bound (>95%) to human plasma proteins. The major active metabolites present in human are the β -hydroxyacid of lovastatin, its 6'-hydroxy derivative, and 2 unidentified metabolites. Peak plasma concentrations of both active and total inhibitors were attained within 2 to 4 hours of dose administration.
3. With a once-a-day dosing regimen, plasma concentrations of total inhibitors over a dosing interval achieved a steady state between the second and third days of therapy and were about 1.5 times those following a single dose.

Indication(s):

For the reduction of elevated total and LDL Cholesterol levels in patients with primary hypercholesterolemia (Type IIa and IIb), when the response to diet and other nonpharmacological measures alone has been inadequate. Lovastatin may be useful to reduce elevated LDL cholesterol levels in patients with combined hypercholesterolemia and hypertriglyceridemia.

Dosage and Administration:

The patient should be placed on a Standard cholesterol-lowering diet before receiving lovastatin and should continue on this diet during treatment with the drug. The usual starting dose is 20mg per day given as a single dose with the evening meal. Adjustment of dosage, if required, should be made at intervals of not less than 4 weeks, to a maximum of 80mg daily given in single doses or divided doses with morning and evening meals.

Concomitant therapy: Preliminary evidence suggests that the cholesterol-lowering effects of lovastatin and the bile acid sequestrant, cholestyramine, are additive.

The combined use of lovastatin with gemfibrozil should be avoided.

In patients taking danazol, verapamil, diltiazem, fibrates (except gemfibrozil) or lipid-lowering dose of niacin (≥ 1 g/day) concomitantly with lovastatin, the dose of lovastatin should not exceed 20mg/day.

In patients taking amiodarone concomitantly with lovastatin, the dose of lovastatin should not exceed 40mg/day.

Dosage in patients with renal insufficiency: In patient with severe renal insufficiency (creatinine clearance <30mL/min) dosage increases above 20mg/day should be carefully considered and, if deemed necessary, implemented cautiously.

To be dispensed on physician's prescription.

Route of Administration: Oral

Contraindication(s):

1. Hypersensitivity to any component of this medication.
2. Active disease or unexplained persistent elevations of serum transaminases.
3. Pregnancy and lactation.
4. Concomitant administration of strong CYP3A4 inhibitors (e.g. itraconazole, ketoconazole, posaconazole, voriconazole, HIV protease inhibitors, boceprevir, telaprevir, erythromycin, clarithromycin, telithromycin and nefazodone).
5. Concomitant administration of cyclosporine.

Precaution(s) / Warning(s):

1. In controlled clinical trials, marked persistent increases (to more than 3 times the upper limit of normal) in serum transaminases occurred in 1.6% of adult patient who received lovastatin for at least 1 year. The increase usually appeared 3-12 months after the start of therapy. It is recommended that liver function tests be performed at baseline and every 4-6 weeks during the first 15 months of therapy with lovastatin and periodically thereafter in all patients. The drug should be used with caution in patients with past history of liver disease.
2. Rhabdomyolysis has occurred rarely and should be considered in any patient with diffuse myalgias, muscle tenderness and/or marked elevation of creatine phosphokinase (10 times the upper limit of normal).
3. Most of the patients who have developed myopathy were receiving concomitant therapy with immunosuppressive drugs included cyclosporine, gemfibrozil or lipid-lowering doses of niacin.
4. Patients should be advised to report promptly unexplained muscle pain, tenderness, weakness particularly if accompanied by malaise or fever.
5. Pediatric use: Safety and effectiveness in children have not been established. Treatment of children with lovastatin is not recommended.
6. Homozygous familial hypercholesterolemia: Lovastatin is less effective in patients with the rare homozygous familial hypercholesterolemia and appears to be more likely to raise serum transaminase in this group of patients.
7. Eye: It is recommended that patients placed on lovastatin therapy be examined with a slit lamp before or shortly after initiation of treatment, and annually thereafter as new opacities was noted during clinical trial. However, this finding is inconclusive at the moment.
8. Colchicine: Cases of myopathy, including rhabdomyolysis, have been reported with lovastatin coadministered with colchicine, and caution should be exercised when prescribing lovastatin with colchicine.

9. There have been very rare reports of an immune-mediated necrotizing myopathy (IMNM) during or after treatment with some statins. IMNM is clinically characterized by:
 - persistent proximal muscle weakness and elevated serum creatine kinase, which persist despite discontinuation of statin treatment;
 - muscle biopsy showing necrotizing myopathy without significant inflammation;
 - improvement with immunosuppressive agents.
10. Myasthenia Gravis/ Ocular Myasthenia: In few cases, statins have been reported to induce de novo or aggravate pre-existing myasthenia gravis or ocular myasthenia. Lovastatin® Tablet should be discontinued in case of aggravation of symptoms. Recurrences when the same or a different statin was (re-) administered have been reported.

Interaction with Other Medicaments:

Contraindicated Drugs

1. Strong inhibitors of CYP3A4: Concomitant use with strong CYP3A4 inhibitors (e.g. itraconazole, ketoconazole, posaconazole, voriconazole, HIV protease inhibitors, boceprevir, telaprevir, erythromycin, clarithromycin, telithromycin and nefazodone) is contraindicated.
2. Cyclosporine: The risk of myopathy/rhabdomyolysis is increased by concomitant administration of cyclosporine. Concomitant use of this drug with lovastatin is contraindicated.

Other Drugs

1. Gemfibrozil, other fibrates, and niacin $\geq 1\text{g/day}$: These drugs increase the risk of myopathy when given concomitantly with lovastatin, probably because they can produce myopathy when given alone. There is no evidence to suggest that these agents affect the pharmacokinetics of lovastatin. Myopathy, including rhabdomyolysis, has occurred in patients who were receiving coadministration of lovastatin with fibric acid derivatives or niacin. Concurrent use of fibrates may cause severe myositis and myoglobinuria.
2. Danazol, verapamil, diltiazem: The risk of myopathy/rhabdomyolysis, is increased by concomitant administration of danazol, verapamil, or diltiazem particularly with higher doses of lovastatin.
3. Amiodarone: The risk of myopathy/rhabdomyolysis is increased when amiodarone is used concomitantly with higher doses of closely related member of HMG-CoA reductase inhibitor class.
4. Colchicine: Cases of myopathy, including rhabdomyolysis, have been reported with lovastatin coadministered with colchicine, and caution should be exercised when prescribing lovastatin with colchicine.
5. When lovastatin and coumarin anticoagulants are administered concomitantly, prothrombin time may be increased in some patients. In patients taking anticoagulants, prothrombin time should be determined prior to starting therapy with lovastatin and thereafter, monitored at the intervals usually recommended for patients on coumarin anticoagulants.

Side Effect(s) / Adverse Reaction(s):

Gastrointestinal	: Constipation, diarrhea, dyspepsia, flatus, abdominal pain/cramps, nausea, heartburn.
Musculoskeletal	: Muscle cramps, myalgia, myopathy, rhabdomyolysis, immune-mediated necrotizing myopathy (frequency not known).
CNS	: Dizziness, headache, fatigue, sleep disorders, insomnia, myasthenia gravis (frequency not known).
Skin	: Rash, pruritus.
Special senses	: Blurred vision, dysgeusia.
Laboratory test	: Marked persistent increases of serum transaminases have been noted.
Eye disorders	: ocular myasthenia (frequency not known).

There have been rare post-marketing reports of cognitive impairment (e.g. memory loss, forgetfulness, amnesia, memory impairment, confusion) associated with statin use. These cognitive issues have been reported for all statins. The reports are generally non-serious and reversible upon statin discontinuation, with variable times to symptom onset (1 day to years) and symptom resolution (median 3 weeks).

Increases in HbA1C and fasting blood glucose have been reported with statins. The risk of hyperglycemia, however, is outweighed by the reduction in vascular risk with statins.

Symptoms and Treatment for Overdosage and Antidote(s):

No specific symptoms and all patients recovered without sequelae in a few reported cases of accidental overdosage. The maximum dosage taken was fifty-two 20mg tablets (1.04g).

In the event of overdosage, treatment should be symptomatic and supportive, liver function should be monitored, and appropriate therapy should be instituted. Until further experience is obtained, no specific therapy of overdosage can be recommended.

Shelf-Life:

2 years from the date of manufacture.

Storage Condition(s):

Keep in a tight container. Store at temperature below 30°C. Protect from light and moisture.

Product Description & Packing(s):

A light blue octagonal tablet, one side impressed with the mark "YUNG SHIN".

Plastic bottle of 1000's (for export only).

Blister packing of 10's x 10 and 10's x 50.



Manufacturer and Product Registration Holder:
Y.S.P. INDUSTRIES (M) SDN. BHD. (199001001034)
 Lot 3, 5, 7, 12 & 14, Jalan P/7, Section 13,
 Kawasan Perindustrian Bandar Baru Bangi,
 43000 Kajang, Selangor, Malaysia.
 Ordering Line: 1 800 88 3027
 Product Info: 1 800 88 3679